

DERMAAI: Explainable 7-Class Dermoscopic Image Classification

A Research Demonstrator Using a Frozen Probability Ensemble & Dual CBAM Attention

1. RESEARCH OBJECTIVE

Skin lesion classification is impeded by severe class imbalance (HAM10000: nv >6:1 vs mel). We systematically evaluated 15 controlled configurations to construct a balanced, explainable ensemble.

2. DATASET SPLIT (Seed=42)

- Train: 8,017 | Val: 1,010 | Held-Out Test: 988
- Strict population isolation maintained throughout all phases.

3. FROZEN ARCHITECTURE

- MobileNetV3-Large + CBAM Attention Head
- 4,326,297 parameters per model (0 trainable)
- Fast inference (~15ms on CPU)

4. FINAL PROBABILITY ENSEMBLE

$P_{\text{final}} = 0.85 P_8 + 0.15 P_{13}$

- Exp 8: Baseline balanced weighting ($\alpha=0.75$)
- Exp 13: Targeted 1.20x melanoma weight
- Calibrated strictly on N=1,010 validation images

5. FROZEN TEST BENCHMARK (N=988)

Accuracy: 80.57% | Balanced Acc: 75.69%

Macro F1: 70.51% | Weighted F1: 81.06%

Melanoma Recall: 50.94% | Mel F1: 52.43%

Project benchmark on N=988 HAM10000 test split. Not clinical accuracy.

6. MELANOMA ERROR ANALYSIS

- mel -> nv = 26 (subtle border/superficial lesions)
- mel -> bkl = 19 (keratotic morphological overlap)
- nv -> mel = 38 (dysplastic/atypical benign nevi)

7. GRAD-CAM ATTENTION

Hooks onto *model.attention* (1x960x7x7). Compares global boundary context (Exp8) with focal pigment networks (Exp13). Interpretability aid, not clinical causality.

8. INTERACTIVE DEMO

- FastAPI Backend + Next.js 16 (React 19)
- Docker containerized, in-memory image validation

9. SCIENTIFIC LIMITATIONS

- Retrospective dataset (no multi-center external validation)
- Repeated test-set evaluation across 15 experiments
- No prospective clinical trial validation

10. MANDATORY DISCLAIMER

IMPORTANT: These results are image-level research benchmarks on the project's HAM10000 evaluation split and do not establish clinical diagnostic performance. RESEARCH USE ONLY.



GitHub Repo



Web Demo